COD returns: a controlled, customer-aware pilot

TO: Founder | FROM: Assignment analysis | DATE: 30 October 2026

Recommendation. Pilot a four-action policy that reserves HOLD for very high risk or clear data-quality overrides, uses confirmation and prepaid nudges for middle-risk orders, and keeps a higher HOLD bar for first-time customers. Do not block COD by pincode alone. The retrospective results are estimates, so start with a measured pilot rather than automatic cancellation.

What the sample says

565 / 1,987	28.4%	169.40
RTO orders / labeled sample	overall historical RTO rate	estimated cost per RTO

Finding	Evidence
First-time customers carry more RTO risk.	First-time: 452/1,495 (30.2%); repeat: 113/492 (23.0%). That argues for reducing friction for new buyers, not applying the same cancellation bar to everyone.
Risk varies sharply by geography.	Bihar: 78/145 (53.8%); Maharashtra: 70/385 (18.2%). Use this gap to focus operations and confirmation, not as a blanket pincode ban.
Text-quality flags are useful, but some groups are small.	Junk-address orders: 91/141 (64.5%); fake-name orders: 21/28 (75.0%). The fake-name group is small, so treat it as a review signal rather than proof.
Acquisition channel matters for new buyers.	First-time Meta orders: 191/563 (33.9%). These orders combine elevated return risk with the stated 300+ acquisition cost.

The risk model's cross-validated AUC is 0.72; on the held-out September 2026 sample it is 0.75. It is useful for ranking orders, but not accurate enough to justify automatic cancellation by itself. Segment rates are historical associations, not causal effects.

Recommended policy

Keep the current customer-specific HOLD thresholds: 85% for first-time customers and 80% for repeat customers. Use CONFIRM and NUDGE below those thresholds; hold only for the high-risk band or clear safety/data-quality overrides. Let geography inform outreach and staffing, not automatic COD denial. Show a plain-language reason and route ambiguous cases to a person.

What the policy would have done last month

The held-out September 2026 sample contains 329 orders. The table compares current rules with an aggressive operations-style proposal and a scenario using the same risk bands for first-time and repeat customers. All rupee results are retrospective estimates under configured assumptions.

Policy	SHIP / CONFIRM / NUDGE / HOLD	First-time held	Estimated net
Current rules	178 / 90 / 60 / 1	1	63
Ops-style proposal	234 / 0 / 0 / 95	75	-29,989
Same bands for all	122 / 146 / 58 / 3	3	244

Balanced rules: 178 SHIP, 90 CONFIRM, 60 NUDGE, and 1 HOLD. Estimated savings of 2,212 less 1,870 expected genuine-order value, 279 discount and message costs gives 63 net. The backtest estimates 13.1 RTOs avoided and 2.1 genuine orders lost on an expected-value basis.

Operations-only pressure is costly: the aggressive proposal held 95 orders, including 75 first-time orders. In this labeled sample, 49 held orders were delivered; the estimated at-stake value was 37,781, against 7,792 in avoided RTO costs, for -29,989 net. That is the growth cost the policy must guard against.

Marketing trade-off: the same-bands scenario estimates 244 net versus 63 for customer-specific rules, but holds 3 first-time orders versus 1. The sample is too small to declare the higher point estimate a winner. Keep first-time protection as the pilot baseline and test the alternative in a controlled group.

Prepaid offer maths

For a repeat order worth 649 at 50% risk, the generated table estimates 59.62 in COD-to-prepaid return-cost savings after gateway fees. The recommended offer is 30 (about 4.6% of order value), leaving 29.62 per conversion under the table's assumptions. Estimated conversion is 7.0%; discount only where expected benefit remains positive.

What could go wrong, and what to build next

The sample is smaller than stated monthly volume, local risk rates can move, and cost/response assumptions may be wrong. A model can also flag a genuine customer or a text provider can fall back. During a two-week pilot, review held orders before cancellation and track action mix, confirmed delivery, RTO, prepaid conversion, discounts, complaints, and realized shipping costs by first-time status and geography. Compare against a randomized control group. Next, connect real WhatsApp and fulfillment events, measure actual outcomes, recalibrate on newer data, and adjust thresholds only when measured contribution supports it.

Decision: pilot the balanced rules, preserve customer choice, and expand only if measured net contribution improves without an unacceptable rise in genuine-order cancellations.